25901439 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY200 - SEMICONDUCTOR ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. Distinguish between JFET and BJT based on their input impedance and noise level? [U] / [CO1]
2. Define transfer characteristics of a DE-MOSFET? [U] / [CO2]
3. List the different types of ICs based on scale of integration. [K] / [CO3]
4. What is meant by the term "summing amplifier" in op-amp applications? [K] / [CO4]
5. Why does an ideal op-amp draw no current at its input terminals, particularly at the summing junction? [U] / [CO4]
6. In which op-amp configuration the phase difference between input and output is out of phase? [K] / [CO4]
7. Define an operational amplifier (op-amp). Why is it called so? [U] / [CO4]
8. Define an op-amp differentiator. Draw its basic circuit diagram. [K] / [CO4]
9. Which are the three basic gates? Give their symbols. [K] / [CO5]

10. Give the truth table and symbol of XOR gate. [K] / [CO5]

11. Write the 1's complement of 1101101_2. [U] / [CO5]

12. Write the output of a NOR gate when both inputs are 0. [U] / [CO6]

13. Simplify the Boolean expression $(A+B)(A+C)$ [A] / [CO6]

14. Illustrate De Morgan's first law using basic gates. [An] / [CO6]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. Describe how reverse bias at the gate controls the drain current in JFET. [U] / [CO1]

16. Explain the effect of positive and negative gate voltage on depletion MOSFET operation. [U] / [CO2]

17. Explain the construction and working of n-channel enhancement only MOSFET [U] / [CO2]

18. Explain 2's complement subtraction using suitable examples. [U] / [CO5]

19. Give a brief explanation of digital integrated circuits, highlighting their practical applications. [U] / [CO3]

20. Add 34 and 89 after converting to binary equivalent. Check your answer in decimal system. [U] / [CO5]

21. What are monolithic ICs? Explain the fabrication of transistors and capacitors in a monolithic IC. [U] / [CO3]

22. Construct the truth table for the Boolean expression $F=(A+B)(\overline{A}+C)$.Also explain how the output changes for different combinations of input variables. [U] / [CO6]

[5x6 = 30]